

Notes: Kroszner and Strahan (QJE, 1999)

What Drives Deregulation? Economics and Politics of the Relaxation of Bank Branching Restrictions

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1 Big picture

- **Research question.** What drives deregulation? More specifically, why did U.S. states relax restrictions on bank branching at different times during the 1970s, 1980s, and early 1990s?
- **Main idea.** Deregulation is modeled as a political-economy outcome. Some groups gain from removing geographic restrictions on banks, while other groups lose protection and therefore resist reform.
- **Theories compared.** The paper evaluates private-interest, public-interest, and political-institutional theories of regulatory change.
- **Empirical setting.** Bank branching restrictions were set mostly at the state level, and states deregulated at different dates. This creates state-by-year variation that is unusual among major U.S. deregulation episodes.
- **Main empirical object.** The authors estimate a hazard model of the time until a state permits intrastate branching through merger and acquisition. The dependent variable is the duration of regulation.
- **Main finding.** Deregulation occurs later when small banks are stronger and when the insurance industry is larger in states where banks can sell insurance. Deregulation occurs earlier when there are more small, bank-dependent firms and when small banks are financially weaker.
- **Federal validation.** The same political-economy variables help explain House votes on the Wylie–Neal amendment to permit interstate bank branching.
- **Interpretation.** The private-interest theory provides the most compelling overall explanation. Public-interest and political-institutional variables matter in some places, but they do not explain the full pattern as well.
- **Economic takeaway.** Deregulation is not simply the automatic removal of inefficient rules. It is shaped by the relative political strength of winners and losers from reform.

2 Incremental contribution of the paper

- **From regulation to deregulation.** Earlier private-interest work was more successful at explaining why regulation appears than why it disappears. This paper studies regulatory exit directly.
- **Rich cross-state variation.** Most major deregulation episodes were national. Bank branching deregulation occurred state by state, providing both cross-sectional and time-series variation.
- **Direct test of competing theories.** The paper maps specific variables to private-interest, public-interest, and political-institutional predictions, then asks which theory best explains timing.
- **Interest-group detail.** The analysis separates intraindustry rivalry between small and large banks, interindustry rivalry between banks and insurance firms, and borrower interests among small firms.

- **Hazard-model framework.** The paper treats regulation as a duration problem and asks which variables raise or lower the hazard of deregulation.
- **Consistency check using Congress.** The authors test whether the same forces explain a House vote on interstate branching, not just state-level legislative timing.
- **Why the 1970s?** The paper also explains why deregulation began when it did by discussing technological, legal, and financial innovations that weakened the old political equilibrium.
- **Appendix contribution.** The appendix shows that branching through merger and acquisition has the clearest effects on growth, banking structure, and bank efficiency, which justifies focusing on that form of deregulation.

3 Institutional setting and theories

3.1 Origins of geographic banking restrictions

Bank branching restrictions had deep historical roots. States used bank charters and restrictions on entry as a way to raise revenue and create local banking rents. Restrictions on interstate banking prevented out-of-state banks from operating in a state's territory, while intrastate limits protected local banking markets within a state.

- **State revenue motive.** States received charter fees, taxes, and sometimes ownership stakes from banks, creating incentives to restrict banking competition.
- **Local monopoly rents.** By limiting branching, states could create protected local banking markets.
- **Persistence.** Many restrictions remained stable from the Great Depression until the deregulation wave that began in the 1970s.
- **Beneficiaries.** Small local banks benefited because restrictions protected them from competition by larger, more efficient banks.

Interpretation. The historical origin of branching restrictions fits the private-interest view: concentrated beneficiaries support rules whose costs are spread across borrowers and consumers of banking services.

3.2 Forms of branching deregulation

The paper distinguishes several types of bank expansion reform.

- **Multibank holding companies.** States could allow bank holding companies to own several banks, though these banks still operated separately.
- **Intrastate branching through merger and acquisition.** States could allow banks or holding companies to convert acquired banks into branches of a single banking organization.
- **Full statewide branching.** Banks could open new branches anywhere in the state.

- **Interstate banking.** States could allow out-of-state bank holding companies to acquire in-state banks, later culminating in national reform through the Riegle–Neal Act.

Key focus. The main hazard analysis focuses on intrastate branching through merger and acquisition because the appendix shows that this type of reform has the strongest and most consistent effect on real and banking-sector outcomes.

3.3 Private-interest predictions

The private-interest theory says deregulation reflects competition among organized groups.

- **Small banks.** Small banks lose protection from branching deregulation. Reform should occur later where small banks hold a larger share of banking assets.
- **Financially strong small banks.** If small banks are healthy, they have more resources and incentives to defend restrictions. Reform should occur later where small banks have higher capital ratios relative to large banks.
- **Large banks.** Large banks gain from deregulation because they can expand across local markets and acquire smaller institutions.
- **Insurance firms.** Where banks can sell insurance, insurers may oppose branching deregulation because banks could become more effective competitors in insurance distribution.
- **Small firms.** Small, bank-dependent firms may favor deregulation because more competitive banking markets can improve credit availability and lower loan costs.

3.4 Public-interest and political-institutional predictions

The public-interest theory predicts deregulation when social costs of regulation are high. The political-institutional theory emphasizes ideology and institutions.

- **Small bank share.** Public-interest theory could predict earlier deregulation where the protected small-bank sector is large, because the social cost of protection is high. This is the opposite of the private-interest prediction.
- **Insurance rivalry.** If branching restrictions prevent scope economies between banking and insurance, public-interest theory predicts earlier deregulation where the insurance sector is large and banks can sell insurance. This is again opposite to the private-interest prediction.
- **Small firms.** Both private-interest and public-interest theories predict earlier deregulation where many small firms depend on bank credit.
- **Loan interest rates.** Public-interest theory predicts earlier deregulation where loan rates are high because borrowers face larger welfare losses.
- **Bank instability.** Both theories can predict earlier deregulation when bank distress is high: public-interest because diversification may reduce failures, and private-interest because weak banks are less able to lobby.
- **Party politics.** Republicans are usually viewed as more pro-deregulation than Democrats, so Democratic control is expected to delay reform.

4 Data and measurement

4.1 Deregulation timing

The main outcome is the year in which each state permits intrastate branching through merger and acquisition.

- **Sample period.** The hazard analysis covers 1970–1992.
- **States at risk.** States that had already deregulated before 1970 are excluded from the hazard sample. The main hazard sample begins with 39 states at risk.
- **Deregulation events.** Thirty-six of the 39 states deregulate during the sample period.
- **Censored states.** Arkansas, Iowa, and Minnesota had not deregulated by the end of the sample in the main hazard analysis.
- **State-year observations.** Because each state is observed until deregulation or censoring, the main sample contains 637 observations.

4.2 Interest-group variables

The key variables measure the relative strength of groups that gain or lose from branching reform.

- **Small bank asset share.** The share of state banking assets held by banks below the median bank size in that state and year.
- **Small bank financial strength.** The capital-to-assets ratio of small banks minus the capital-to-assets ratio of large banks in the state.
- **Insurance sector size.** Insurance gross state product divided by insurance plus banking gross state product.
- **Bank insurance powers.** An indicator for whether banks in the state may sell insurance.
- **Small firm share.** The share of state establishments with fewer than twenty employees, used as a proxy for small bank-dependent firms.

4.3 Public-interest and political variables

Additional variables capture borrower costs, bank stability, and political institutions.

- **Loan yield spread.** Average yield on bank loans in the state minus the federal funds rate.
- **Bank failure rate.** Assets in failed banks divided by total state bank assets.
- **Democratic control.** The share of the three state-government bodies controlled by Democrats: the assembly, senate, and governorship.
- **One-party control.** Indicator equal to one if the same party controls the governorship and both legislative chambers.

- **Initial restrictions.** Some robustness tests include indicators for unit banking and changes in bank insurance powers.

5 Methodology and empirical specifications

5.1 Hazard model for state deregulation

The empirical design treats regulation as a duration. The hazard rate is the probability that a state deregulates at time t , conditional on not having deregulated earlier.

$$h[t, x(t), \beta] = h_0(t) \exp[x(t)' \beta], \quad h_0(t) = pt^{p-1}.$$

The authors estimate a Weibull proportional-hazards model. The Weibull shape is useful because the nonparametric hazard rises over time, especially in the late 1980s.

The paper reports the results in the log expected time metric:

$$\ln(T) = x' \beta^* + \varepsilon, \quad \beta^* = -\beta/p.$$

- **Positive coefficient.** A positive coefficient means the variable increases expected time to deregulation, so it delays reform.
- **Negative coefficient.** A negative coefficient means the variable reduces expected time to deregulation, so it speeds reform.
- **Economic interpretation.** The authors translate one-standard-deviation changes into changes in years until deregulation, using the mean time to deregulation of 15.8 years.

5.2 Robustness tests

The paper checks whether the main hazard findings reflect omitted initial conditions or regional differences.

- **Loan rates.** Adding average loan yield spreads reduces the sample because these data start later, but it does not change the main interest-group results.
- **Bank failures.** State bank failure rates are not a strong predictor of state-level deregulation timing.
- **Unit banking.** Unit banking states deregulate later, but controlling for unit banking leaves the main coefficients largely intact.
- **Insurance powers.** Controlling for whether a state changed bank insurance powers does not overturn the main results.
- **Regional indicators.** Adding broad regional controls preserves the key small-bank and small-firm effects.

5.3 House vote on interstate branching

The authors also study a federal roll-call vote: the November 14, 1991 Wylie–Neal amendment to permit interstate bank branching.

$$\Pr(\text{Vote}_{rs} = 1) = \Phi(\alpha + X'_s\delta + \theta \text{Democrat}_r).$$

- **Dependent variable.** The dependent variable equals one if the Representative votes for the amendment and zero otherwise.
- **State-level variables.** The explanatory variables are the same political-economy variables used in the hazard model.
- **Party variable.** The state-government political variables are replaced by a legislator-level Democrat indicator.
- **Interpretation.** In this probit setting, a negative coefficient means the variable lowers the probability of voting for deregulation.

6 Identification and interpretation

6.1 What the paper can identify

The paper identifies whether states with different interest-group structures deregulate at different times and whether legislators from states with those same structures vote differently on federal branching reform.

This is not a randomized experiment. The evidence is strongest as a political-economy pattern supported by timing variation, robustness tests, consistency between state and federal evidence, and ex post consequences of deregulation.

6.2 Why small banks are central

Small banks are the clearest protected interest. Branching restrictions shield them from larger banking organizations. The paper’s most important finding is that states deregulate later when small banks hold a larger asset share.

Interpretation. This result is difficult to reconcile with a simple public-interest story, because the public-interest prediction would often be that states with a larger protected sector should deregulate earlier due to larger efficiency costs.

6.3 Insurance as a rival industry

The insurance result is important because it examines rivalry outside banking. Where banks can sell insurance, a larger insurance sector delays deregulation. This is consistent with insurers opposing bank expansion that would make banks stronger competitors in selling insurance products.

Interpretation. This result is especially supportive of private-interest theory because public-interest theory would predict that larger scope economies should make deregulation more socially

valuable.

6.4 Small firms and borrower interests

States with more small firms deregulate earlier. This result can be interpreted in both private-interest and public-interest terms.

- **Private-interest view.** Small, bank-dependent firms are an organized or politically relevant beneficiary of reform.
- **Public-interest view.** The social cost of branching restrictions is higher when more firms depend on local bank credit.

6.5 What the federal vote adds

The House vote is a useful validation exercise. The same small-bank and insurance variables that delay state deregulation also reduce the probability that a Representative votes for interstate branching reform.

Interpretation. This supports the idea that the state-level results are not just artifacts of the hazard model. The same interest-group structure appears in a different political arena.

7 Tables and figures

7.1 Table I and Figure I: Timing and geography of branching deregulation

Read:

- Table I reports when each state permitted intrastate branching through merger and acquisition, full intrastate branching, interstate banking, and multibank holding companies.
- The table shows substantial cross-state variation. Some states had already permitted intrastate branching before 1970, while others waited until the late 1980s or early 1990s.
- Figure I maps the year of intrastate branching deregulation across states.
- The map shows that deregulation was not a single national shock. It spread unevenly across regions and states.

Economic message: The empirical strategy depends on the fact that branching deregulation occurred gradually and unevenly. This variation allows the authors to ask which state characteristics predict earlier or later reform.

TABLE I
YEAR OF DEREGULATION OF RESTRICTIONS ON GEOGRAPHICAL EXPANSION, BY STATE

State	Intrastate branching through M&A	Full intrastate branching permitted	Interstate banking permitted	Multibank holding companies permitted
AL	1981	1990	1987	<1970
AK	<1970	<1970	1982	<1970
AZ	<1970	<1970	1986	<1970
AR	1994	**	1989	1985
CA	<1970	<1970	1987	<1970
CO	1991	**	1988	<1970
CT	1980	1988	1983	<1970
DE	<1970	<1970	1988	<1970
DC	<1970	<1970	1985	<1970
FL	1988	1988	1985	<1970
GA	1983	**	1985	1976
HI	1986	1986	**	<1970
ID	<1970	<1970	1985	<1970
IL	1988	1993	1986	1982
IN	1989	1991	1986	1985
IA	**	**	1991	1984
KS	1987	1990	1992	1985
KY	1990	**	1984	1984
LA	1988	1988	1987	1985
ME	1975	1975	1978	<1970
MD	<1970	<1970	1985	<1970
MA	1984	1984	1983	<1970
MI	1987	1988	1986	1971
MN	1993	**	1986	<1970
MS	1986	1989	1988	1990
MO	1990	1990	1986	<1970
MT	1990	**	1993	<1970
NE	1985	**	1990	1983
NV	<1970	<1970	1985	<1970
NH	1987	1987	1987	<1970
NJ	1977	**	1986	<1970
NM	1991	1991	1989	<1970
NY	1976	1976	1982	1976
NC	<1970	<1970	1985	<1970
ND	1987	**	1991	<1970
OH	1979	1989	1985	<1970
OK	1988	**	1987	1983
OR	1985	1985	1986	<1970
PA	1982	1990	1986	1982
RI	<1970	<1970	1984	<1970
SC	<1970	<1970	1986	<1970
SD	<1970	<1970	1988	<1970
TN	1985	1990	1985	<1970
TX	1988	1988	1987	1970
UT	1981	1981	1984	<1970
VT	1970	1970	1988	<1970
VA	1978	1987	1985	<1970
WA	1985	1985	1987	1981
WV	1987	1987	1988	1982
WI	1990	1990	1987	<1970
WY	1988	**	1987	<1970

**States not yet deregulated. Source: Amel [1993] and updates by authors.

Table I: Year of deregulation of restrictions on geographical expansion, by state.

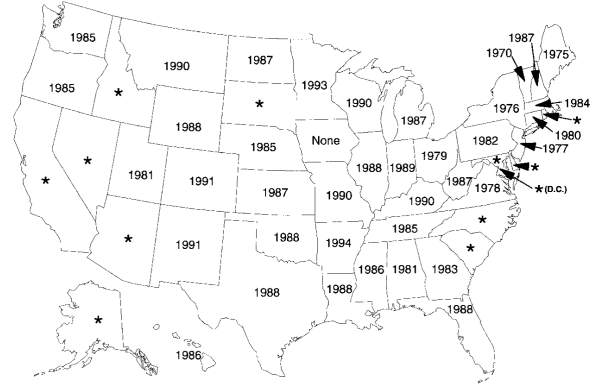


FIGURE I
Deregulation of Restrictions on Intrastate Branching
*Permitted Intrastate Branching before 1970.

Figure I: Deregulation of restrictions on intrastate branching.

7.2 Figure II and Table II: Hazard shape and explanatory variables

Read Figure II:

- Figure II shows the Kaplan–Meier estimate of the hazard rate for branching deregulation.
- The hazard is low and relatively flat in the early part of the sample, then rises substantially in the late 1980s and early 1990s.
- This shape motivates the Weibull hazard specification with positive duration dependence.

Read Table II:

- Table II reports summary statistics and correlations for the main hazard-model variables.
- The average small bank asset share is 0.11, meaning small banks hold about 11 percent of banking assets in the typical state-year observation.

- The small firm share is high on average, 0.88, but varies enough across states to help explain deregulation timing.
- The correlation table shows that small bank asset share is strongly negatively correlated with the relative capital ratio of small banks, which makes robustness checks important.

Economic message: Deregulation becomes more likely over calendar time, but the paper's main question is whether political-economy variables explain which states move earlier within this common deregulation wave.

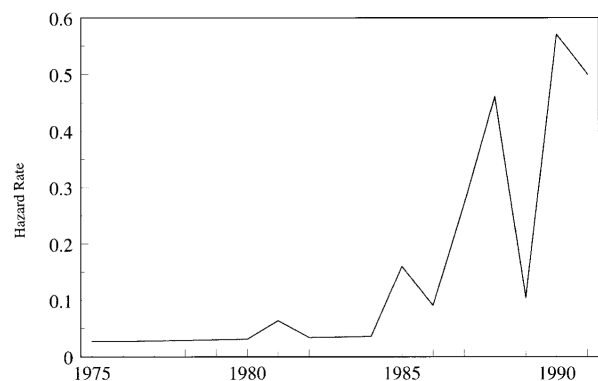


FIGURE II
Kaplan-Meier Estimate of Hazard Rate for Branching Deregulation

Figure II: Kaplan-Meier estimate of hazard rate for branching deregulation.

TABLE II
SUMMARY STATISTICS FOR VARIABLES IN HAZARD MODEL, 1970-1992

Panel A. Univariate statistics	Standard deviation						
	Mean		Minimum	Maximum			
	(1)	(2)	(3)	(4)			
Small bank asset share of all banking assets in state	0.11	0.05	0.01	0.23			
Relative size of insurance to banking plus insurance in the state	0.47	0.09	0.24	0.80			
Indicator is 1 if banks can sell insurance in the state	0.20	0.40	0	1			
Small firm share of the number of firms in the state	0.88	0.02	0.78	0.95			
Capital ratio of small banks relative to large in the state	0.02	0.01	-0.01	0.09			
Share of state government controlled by Democrats	0.65	0.36	0	1			
Indicator is 1 if state controlled by one party	0.54	0.50	0	1			
Average yield on bank loans in the state minus Fed funds rate	0.02	0.02	-0.04	0.08			
Panel B. Correlations	Small bank	Relative size of insurance	Insurance indicator	Small firm	Small bank capital	Dem. control	Single party
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Small bank asset share of all banking assets in state	1					
Relative size of insurance to banking plus insurance in the state	-0.34	1					
Indicator is 1 if banks can sell insurance in the state	0.20	-0.12	1				
Small firm share of the number of firms in the state	0.30	-0.11	0.05	1			
Capital ratio of small banks relative to large in the state	-0.66	0.20	-0.10	-0.06	1		
Share of state government controlled by Democrats	-0.06	0.01	-0.18	-0.18	0.05	1	
Indicator is 1 if state controlled by one party	0.07	0.12	0.01	-0.06	-0.06	0.30	1
Average yield on bank loans in the state minus Fed funds rate	-0.01	-0.15	0.03	0.22	-0.10	-0.03	-0.06

The number of observations is 637, except for the average interest rate on bank loans which becomes available only in 1976, so the number of observations is 408.

Table II: Summary statistics for variables in the hazard model.

7.3 Tables III and IV: Main hazard results and congressional voting

Read Table III:

- Table III is the main state-level hazard result. The dependent variable is log expected time to

branching deregulation.

- Small bank asset share has a positive and highly significant coefficient. States with stronger small-bank interests deregulate later.
- The relative capital ratio of small banks is also positive, meaning financially stronger small banks are associated with delayed deregulation.
- Where banks can sell insurance, a larger insurance sector delays deregulation. This supports the interindustry rivalry interpretation.
- Small firm share is negative and significant, implying that states with more small firms deregulate earlier.
- Democratic control delays deregulation, while one-party control has little stable effect.

Read Table IV:

- Table IV estimates a probit model of House votes on the Wylie–Neal amendment.
- Legislators from states with larger small-bank shares are less likely to vote for interstate branching reform.
- In states where banks may sell insurance, a larger insurance sector also reduces support for reform.
- Democratic Representatives are less likely to vote for the amendment.
- The federal voting evidence mirrors the state-level hazard evidence.

Economic message: The same groups that delay state-level deregulation also oppose federal branching deregulation. This is the strongest evidence that private interests, not only broad efficiency concerns, shape reform.

TABLE III

HAZARD MODEL OF POLITICAL ECONOMY FACTORS AFFECTING THE TIMING OF STATE BRANCHING DEREGULATION, 1970–1992

	(1)	(2)	(3)	(4)	(5)	(6)
Small bank asset share of all banking assets in state	6.48*** (1.22)	5.19*** (1.37)	6.32*** (1.35)	5.16*** (1.18)	4.88*** (1.33)	4.45*** (0.88)
Capital ratio of small banks relative to large in the state	13.25*** (3.48)	9.21*** (3.19)	13.01*** (3.62)	10.67*** (3.46)	11.44*** (3.75)	9.06** (3.81)
Relative size of insurance in states where banks may not sell insurance, 0 otherwise	3.24** (1.61)	2.59* (1.49)	3.15* (1.65)	2.85* (1.54)	2.72* (1.59)	0.55 (1.34)
Indicator is 1 if banks may sell insurance in the state	0.45*** (0.10)	0.37*** (0.11)	0.45*** (0.10)	0.38*** (0.08)	0.36*** (0.09)	0.01 (0.12)
Relative size of insurance in states where banks may not sell insurance, 0 otherwise	-0.93** (0.43)	-0.90*** (0.33)	-1.02** (0.51)	-0.93* (0.48)	-0.96* (0.53)	-0.04 (0.39)
Small firm share of the number of firms in the state	-9.72*** (2.11)	-6.14*** (2.15)	-9.52*** (2.11)	-9.43*** (2.30)	-9.52*** (2.70)	-15.09*** (2.76)
Share of state government controlled by Democrats	0.31** (0.12)	0.23** (0.11)	0.30** (0.12)	0.27** (0.11)	0.26** (0.11)	0.11 (0.13)
Indicator is 1 if state controlled by one party	-0.04 (0.10)	-0.04 (0.07)	-0.04 (0.07)	-0.01 (0.09)	-0.02 (0.10)	0.17* (0.09)
Average yield on bank loans in the state minus Fed funds rate	—	0.23 (3.19)	—	—	—	—
Bank failure rate	—	—	0.92 (1.11)	—	—	—
Indicator is 1 if state has unit banking law	—	—	—	0.18** (0.10)	0.19* (0.10)	0.30*** (0.09)
Indicator is 1 if state changes bank insurance powers	—	—	—	—	-0.08 (0.12)	-0.26** (0.13)
Includes regional indicators?	No	No	No	No	No	Yes
N	637	408	637	637	637	637
Log likelihood	-3.74	9.00	-3.60	-2.12	-1.91	11.92
p-value of χ^2 for regression	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01

The hazard model is Weibull, where the dependent variable is the log expected time to branching deregulation: $\ln(T) = b^*x + e$. All variables are measured for each state in each year. Small bank asset share is the percent of banking assets in the state held by banks below the median size of banks in each state in each year. Relative capital ratio is the capital to assets ratio of small banks minus that of large banks. Size of insurance relative to banking plus insurance in the state is measured as gross state product from insurance divided by gross state product from insurance plus banking. Insurance indicator is 1 if state law permits banks to enter the insurance business. Small firm share is the percent of all establishments in the state that have fewer than twenty employees. Party control variable is the share of the three bodies of state government controlled by Democrats. One party control indicator is 1 if the same party controls the governorship and has majorities in both chambers of the state legislature. Average yield on bank loans equals total interest income on all domestic loans made by banks in the state divided by total loans. Bank failure rate is the ratio of assets in failed banks during a given state and year divided by total bank assets in that state and year. Unit banking indicator equals 1 for states with unit banking restrictions. The change in insurance powers indicator equals one if the state changed the law to permit banks to sell insurance during the sample period. We divide the country into four regions and include regional indicators in column (5). Robust standard errors are in parentheses. ***, **, and * denote statistically significant at the 1, 5, and 10 percent levels.

Table III: Hazard model of political-economy factors affecting state branching deregulation.

TABLE IV
MARGINAL EFFECTS FROM A PROBIT MODEL OF THE INFLUENCE OF POLITICAL ECONOMY FACTORS ON THE HOUSE VOTE ON THE WYLIE-NEAL AMENDMENT TO PERMIT INTERSTATE BANK BRANCHING, NOVEMBER 14, 1991

	(1)	(2)	(3)	(4)	(5)	(6)
Small bank asset share of all banking assets in state	-4.17*** (1.24)	-4.54*** (1.22)	-4.07*** (1.20)	-4.43*** (1.47)	-4.50*** (1.49)	-4.90*** (1.47)
Capital ratio of small banks relative to large in the state	1.80 (3.75)	2.68 (3.42)	2.56 (3.75)	1.69 (3.78)	1.53 (3.80)	2.45 (4.25)
Relative size of insurance in states where banks may sell insurance, 0 otherwise	-2.34** (1.02)	-2.49** (1.14)	-2.33** (1.01)	-2.49** (1.04)	-2.51** (1.02)	-1.54 (1.11)
Indicator is 1 if banks may sell insurance in the state	0.79*** (0.06)	0.74*** (0.06)	0.69*** (0.06)	0.70*** (0.06)	0.70*** (0.06)	0.67*** (0.07)
Relative size of insurance in states where banks may not sell insurance, 0 otherwise	1.80*** (0.66)	2.42*** (0.72)	1.54** (0.67)	1.73*** (0.66)	1.72*** (0.67)	2.03*** (0.91)
Small firm share of the number of firms in the state	-1.23 (0.92)	-0.83 (0.89)	-1.62 (0.97)	-1.32 (0.95)	-1.30 (1.00)	-0.94 (1.16)
Indicator is 1 if Representative is a Democrat	-0.35*** (0.07)	-0.35*** (0.07)	-0.36*** (0.07)	-0.35*** (0.07)	-0.35*** (0.07)	-0.36*** (0.07)
Average yield on bank loans in the state minus Fed funds rate	—	10.01* (5.15)	—	—	—	—
Average failure rate, 1985 to 1991	—	—	2.28 (1.41)	—	—	—
Indicator is 1 if state has unit banking law	—	—	—	0.04 (0.07)	0.04 (0.07)	-0.01 (0.08)
Indicator is 1 if state changes bank insurance powers	—	—	—	—	-0.01 (0.07)	-0.03 (0.07)
Includes regional indicators?	No	No	No	No	No	Yes
Pseudo- R^2	0.17	0.18	0.17	0.17	0.17	0.18
Log likelihood	-240.24	-238.26	-239.51	-240.13	-240.10	-238.08
p-value of χ^2 for regression	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01

The dependent variable is one if the legislature votes for the amendment and zero if against. The reported coefficients are the effects of a unit change of the independent variable from the mean on the probability of voting in favor of the amendment. All variables are measured by state. $N = 418$, the number of Representatives voting on the amendment. Small bank asset share is the percent of banking assets in the state held by banks below the median size of bank in each state in each year. Relative capital ratio is the capital to assets ratio of small banks minus that of large banks. Size of insurance relative to banking plus insurance in the state is measured as gross state product from insurance divided by gross state product from insurance plus banking. Indicator is one if state law permits banks to enter the insurance business. Small firm share is the percent of all establishments in the state that have fewer than twenty employees. Party indicator is one if the Representative is a Democrat. Average yield on bank loans equals total interest income on all domestic loans made by banks in the state divided by total loans. The average bank failure rate is the ratio of assets in failed banks during a given state divided by total bank assets in that state averaged over the years 1985 to 1991. Unit banking indicator equals one for states with unit banking restrictions. The change in insurance powers indicator equals one if the state changed the law to permit banks to sell insurance during the sample period. We divide the country into four regions and include regional indicators in column (5). Robust standard errors are in parentheses. ***, **, and * denote statistically significant at the 1, 5, and 10 percent levels.

Table IV: Probit model of the House vote on the Wylie–Neal amendment.

7.4 Tables V and VI: Why deregulation began and why M&A branching is the focus

Read Table V:

- Table V reports broad trends in commercial banking from 1950 to 1995.
- ATMs rise from zero before 1975 to more than 120,000 by 1995.
- Money market mutual funds grow from essentially zero to hundreds of billions of dollars.
- The banking system's share of deposits plus money-market funds falls, and small banks' share of banking assets declines.
- Bank failures rise sharply in the 1980s, increasing awareness of the cost of geographically undiversified banks.

Read Table VI:

- Table VI compares the effects of different forms of deregulation on state economic growth, banking structure, and bank efficiency.
- Branching through merger and acquisition is significant and has the expected sign across all five outcomes.
- This form of reform is associated with faster personal-income growth, fewer banking companies, more branches per banking company, lower noninterest expenses, and lower loan-loss provisions.
- Other deregulation indicators have less consistent effects.

Economic message: Table V explains why the old political equilibrium weakened in the 1970s and 1980s. Table VI explains why the paper focuses on intrastate branching through merger and acquisition as the economically important deregulation event.

TABLE V
BROAD TRENDS IN COMMERCIAL BANKING, 1950–1995

Year	Number of ATMs	Domestic bank deposits (billions)	Money market mutual fund (billions)	Percent of deposits + money funds held by banks	Small banks' percent of banking assets	Average number of bank failures
	(1)	(2)	(3)	(4)	(5)	(6)
1950	0	\$ 154	\$ 0	100	NA	4
1955	0	191	0	100	NA	3
1960	0	228	0	100	24	2
1965	0	330	0	100	20	4
1970	0	479	0	100	18	6
1975	9,750	775	4	99	18	6
1980	18,500	1,182	76	94	17	10
1985	61,117	1,787	242	88	14	60
1990	80,156	2,339	493	83	11	179
1995	122,706	2,552	745	77	8	61

Column (1): ATM figures are from Bank Network News, The EFT Network Data Book (New York: Faulkner and Gray, Inc.). The 1975 figure was unavailable. 9,750 is the number of ATMs in 1978, the first year for which complete data are available. *Columns (2)–(4):* Banks' domestic deposits are from the Reports of Income and Condition; money market mutual funds are from the Flow of Funds. Data on all bank deposits, foreign plus domestic are only available beginning in 1970. The trend in banks' share (column (4)) is the same using total deposits instead of domestic deposits. *Column (5):* Percent of banking assets held by small banks, where a small bank is defined as a commercial bank with less than \$100 million in assets in 1994 dollars. These data are based on the Reports of Income and Condition. Data on small banks are not available before 1960. *Column (6):* Five-year average number of bank failures, where the final year is indicated in the first column. These data are from FDIC, Annual Report and the Quarterly Banking Profile.

Table V: Broad trends in commercial banking, 1950–1995.

TABLE VI
MULTIVARIATE REGRESSIONS OF THE EFFECTS OF DIFFERENT TYPES OF BRANCHING
DEREGULATION ON ECONOMIC GROWTH, BANKING STRUCTURE,
AND BANK EFFICIENCY

	<i>Dependent variable</i>				
	Growth in personal income	Log of # of banking companies	Branches per banking company	Log of non-interest expenses	Loan loss provisions/ (loans + leases)
Merger & acquisition branching indicator	0.013*** (0.004)	−0.162*** (0.037)	1.146** (0.453)	−0.107*** (0.035)	−0.007*** (0.001)
Unrestricted state-wide branching indicator	−0.008 (0.005)	0.086 (0.059)	−0.043 (0.574)	0.070* (0.037)	0.005** (0.002)
Interstate banking indicator	0.006 (0.005)	−0.052 (0.057)	1.064** (0.511)	0.005 (0.030)	−0.001 (0.001)
Multibank holding company indicator	−0.005 (0.006)	−0.096* (0.052)	−0.364 (0.529)	−0.095** (0.036)	−0.001 (0.001)
Dependent variable mean [median]	0.086 [0.082]	4.881 [5.136]	6.649 [5.055]	13.812 [13.803]	0.009 [0.006]

This table contains regressions of state-level economic growth and state-level measures of the structure and efficiency of the banking industry. Each dependent variable is regressed on a set of state indicator variables, a set of year indicator variables, and the set of four deregulation indicators; these deregulation indicators are equal to one after deregulation and zero before. If a state was always regulated, the indicator is zero throughout; if it was always deregulated, the indicator is one throughout. Sample period is 1975 to 1991, and Delaware and South Dakota are dropped, so $N = 833$. Noninterest expenses become available only in 1984, so $N = 392$. The coefficient on the deregulation indicators is reported, with robust standard errors in parentheses. ***, **, and * denote statistically significant at the 1, 5, and 10 percent levels.

Table VI: Effects of different types of branching deregulation.

8 Short summary

- The paper studies why U.S. states relaxed bank branching restrictions at different times.
- It compares private-interest, public-interest, and political-institutional theories of deregulation.
- The main empirical method is a Weibull hazard model of the time until intrastate branching through merger and acquisition.
- States deregulate later when small banks hold a larger share of banking assets.
- States also deregulate later when small banks are financially stronger.

- Where banks can sell insurance, a larger insurance sector delays deregulation.
- States with more small firms deregulate earlier, consistent with borrower interests favoring reform.
- Democratic control of state government tends to delay deregulation.
- The same interest-group variables explain a House vote on interstate branching reform.
- Broad technological and financial innovations in the 1970s and 1980s weakened the old equilibrium that supported branching restrictions.
- Overall, the private-interest theory provides the most compelling explanation of the timing of deregulation.

9 Conclusion

9.1 Core conclusion

Kroszner and Strahan show that the deregulation of bank branching restrictions is best understood as a political-economy process. Deregulation happens when the relative political strength of groups favoring reform rises and the relative strength of protected groups falls.

9.2 Economic intuition

Branching restrictions protected small local banks from competition by larger banks. As technology, financial innovation, and legal change reduced the value of local banking monopolies, the coalition defending restrictions weakened. At the same time, large banks and small bank-dependent firms had incentives to support reform.

9.3 Private interest versus public interest

Some results are consistent with public-interest logic, especially the finding that deregulation occurs earlier where many small firms may benefit from more competitive banking. However, the central small-bank and insurance results are difficult to explain with public-interest theory alone. They are much more naturally interpreted as evidence of protected incumbent groups resisting deregulation.

9.4 Implications

The paper implies that regulatory reform depends not only on efficiency gains but also on the distribution of gains and losses. Policy changes become politically feasible when the beneficiaries of existing rules lose power or when the beneficiaries of reform become stronger.

9.5 Limitations

The design is observational, and unobserved state-level political or economic factors could be correlated with both interest-group variables and deregulation timing. The authors address this concern

with robustness checks, federal voting evidence, and ex post consequences of deregulation, but the evidence should still be interpreted as political-economy evidence rather than a randomized causal estimate.

9.6 Takeaway

The enduring takeaway is that deregulation is itself endogenous. Bank branching reform did not occur simply because restrictions were inefficient; it occurred when economic, technological, and political forces changed the balance of power among the groups that benefited from or were harmed by the restrictions.